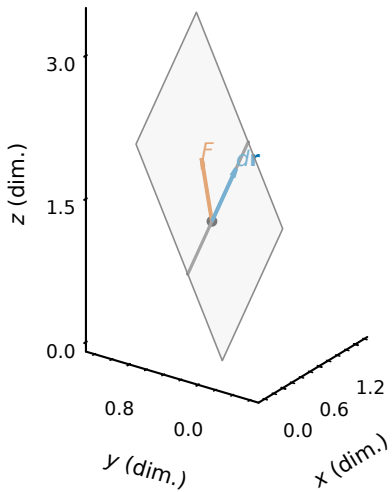
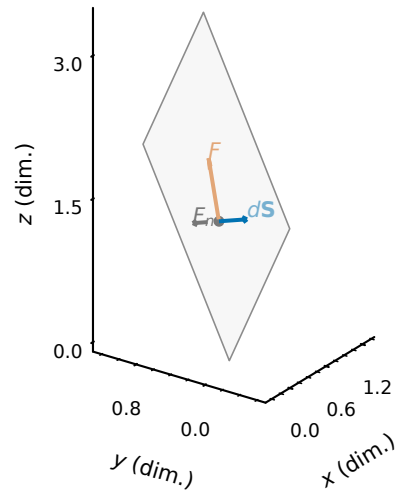
A



$ds = \sqrt{2}\,du;\ F \cdot d\mathbf{r} = 2du$

B



$dS = \sqrt{6}\,du\,dv;\ F \cdot d\mathbf{S} = -2du\,dv$